

# Four Languages, One Structure

## *The Computational Grounding of $\perp$*

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This companion explains the ideas in plain language. It is not the formal document — every claim here restates a result already proved in the technical document ZP-K Computational Grounding. Consult that document for the authoritative mathematics.

### What Is ZP-K Doing?

ZP-J proved that the Quine atom (set-theoretic self-reference) and the bottom element (order-theoretic minimum) are the same object. ZP-K adds a fourth language: computability theory. It adds a fourth description of  $\perp$ , this time in terms of Turing machines and Kleene's second recursion theorem. THREE of the four descriptions are proved to name the same structural role. ZP-K carries  $c_0$ 's computational face as a KleeneStructure requirement (botCode\_is\_quine); reading  $c_0$  as the Kleene quine is that commitment, not a theorem.

On the framework's reading,  $\perp$  is not a description of a Turing machine waiting for an outside party to run it:  $\perp$  is read as an instance of a Turing machine (its ground state, serving as its own program), so if it runs at all, it runs itself (Path 1, the framework's requirement, not a theorem). That rules out an outside executor, not an idle  $\perp$ :  $\perp$  is the unique self-containing state of MachinePhase (da1\_closed\_concrete) even in a dynamics where nothing ever leaves  $c_0$  (tsnap\_holds\_but\_nothing\_moves). That  $\perp$  does run is the occurrence commitment, which DA-1 uses and does not supply. Three of the four descriptions of  $\perp$  are proved to name the same structural object; the computational one is assumed rather than derived.

### What Is a Kleene Fixed Point?

Kleene's second recursion theorem is a foundational result in computability theory. Informally, it says: for any way of transforming programs, there exists a program that is a fixed point of that transformation — a program whose behavior is the same before and after the transformation is applied.

A Kleene fixed point of self-application is called a computational Quine: a program  $c$  such that running  $c$  produces the same output as running  $c$  on its own source code. The definition says only that: it fixes what  $c$  does, and it says nothing about minimality — there is no claim here that no shorter program generates  $c$ , which would need Kolmogorov complexity and plays no formal part in this document.

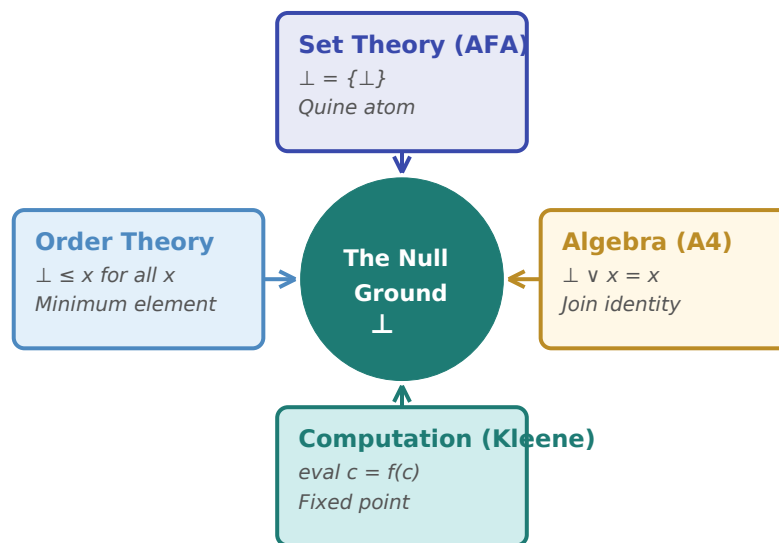
This is not a running computation that might loop forever. Kleene's theorem is an existence proof — it guarantees the fixed point exists before any execution takes place, by a direct construction, not by iterating toward a limit. The question "will it halt?" does not arise: the fixed point is identified by the theorem itself, not discovered by running a potentially divergent process.

**Real-world analogy — A self-printing program**

A Quine program in computer science is a program that, when run, outputs its own source code. It needs no external file to read — the source is baked in. Kleene's theorem guarantees that for any computable transformation, such a fixed point always exists. On the framework's reading, the computational Quine is the expression of  $\perp = \{\perp\}$  in the language of programs:  $c$  is its own program, just as  $\perp$  is its own member. ZP-K carries that reading as a KleeneStructure requirement (botCode\_is\_quine), not a theorem.

## T-COMP: The Three-Way Equivalence, and What Computation Adds

ZP-K's central theorem, T-COMP (Computational Grounding), proves that THREE of the four descriptions below are equivalent - self-containment, order, and algebra all identify the same object. The fourth, the computational one, is not a clause of the theorem: it is supplied as an assumption when the structure is built. The framework holds to that distinction:



The null ground element  $\perp$  at the center, described in four formal languages. The arrows among the first three are proved equivalences. The computational description is joined to them by assumption, not by proof.

| Language             | Property                                           | Intuition                                                                                                                       |
|----------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Set theory (AFA)     | $\perp = \{\perp\}$ — Quine atom                   | $\perp$ is its own sole member; no external interpreter exists                                                                  |
| Order theory (ZP-A)  | $\perp \leq x$ for all $x$ — minimum               | $\perp$ is below everything; the universal starting point                                                                       |
| Algebra (ZP-A A4)    | $\perp \vee x = x$ for all $x$ — join identity     | $\perp$ contributes nothing to any join; the additive zero                                                                      |
| Computation (Kleene) | $eval\ botCode = selfApply\ botCode$ — fixed point | a fixed point of self-application; reading it as " $\perp$ is its own program" is the framework's commitment, not a consequence |

The framework READS the four descriptions as projections of a single structural identity rather than as independent convergences - that the Kleene fixed-point property and the AFA Quine atom property are one property in two vocabularies. That reading is the motivation for building a KleeneStructure, the formal bridge connecting the set-theoretic (AFAStructure) and computational (Kleene fixed-point) descriptions. It is not something ZP-K derives: the bridge is built because the reading is held, not the other way round.

Remember: T-COMP is not saying that set theory and computability theory are the same. It is saying that the specific structural role played by  $\perp$  — self-containment, minimality, join-identity — is expressible in several languages, and that THREE of those expressions are provably equivalent; the computational one is joined by assumption. The equivalence is local to this structural role, not a global identification of the frameworks.

## DA-1: what Lean witnesses

DA-1 (Instantiation as Execution) had three informal argument paths in ZP-E:

Path 1 (Structural — AFA): Nothing external to  $\perp$  can execute  $\perp$ , so if  $\perp$  executes at all, it executes itself. That rules out an outside executor; it does not rule out an inert  $\perp$ . ZP-J proved axiom-free that  $\perp$  is the unique self-containing element (the structural fixed point); that this is the literal  $\perp = \{\perp\}$  holds in the ZF+AFA setting. What Lean witnesses via ZP-K: Path 1's Lean counterpart is `da1_closed_concrete`, which proves `IsQuineAtom ( $\perp$  : MachinePhase)` from the AFAStructure instance `machinePhaseAFA`. There `selfMem` is defined as  $x = \perp$ , so the content is the uniqueness rather than the membership.

Path 2 (Informational — L-INF): The surprisal of  $\perp$  is unbounded — no finite interpreter can hold it as a stored description. Remains outside Lean scope: the step from "unbounded surprisal" to "executing" is a bridge principle of its own, which the framework does not adopt, and it is not the occurrence commitment.

Path 3 (Computational — Kleene): No shorter program generates  $\perp$ . That rules out a shorter outside generator, not an inert string, so Path 3 shows that executing is not derivable from incompressibility. The reading that  $\perp$  is its own program — a Kleene fixed point of self-application — is not a Lean result. ZP-K supplies `botCode_is_quine` as a requirement of the KleeneStructure class, assumed when the structure is built rather than derived; and the condition it asks for is a periodicity condition that even a constant code meets. The "no shorter program" reading needs Kolmogorov complexity, which is uncomputable and plays no formal part here. None of the three paths derives what DA-1 needs: that the configuration reaching  $P_0$  is running is the occurrence commitment, which DA-1 uses and does not supply (ZP-E).

`da1_closed_concrete` (Kleene.lean)

`IsQuineAtom( $\perp$  : MachinePhase)` — proved in Lean 4. The initial machine state  $c_0$  is self-containing. The Lean proves nothing about execution: that the configuration reaching  $P_0$  is running, not an inert description, is the framework's occurrence commitment (ZP-E). DA-1 Path 1 (structural): its Lean counterpart is proved, and Path 1 rules out an external executor, not an inert  $\perp$ . Path 3: its Lean counterpart is a KleeneStructure assumption, not a derivation. Path 2: outside Lean scope (an unadopted bridge principle).

What does it mean, on the framework's reading, that  $\perp$  IS an instance of a Turing machine in its ground state? In ZP-C, the model distinguishes  $c_0$  (the initial configuration, before any instruction executes) from  $c_1$  (after the first instruction fetch). DP-2 (ZP-E) proved that these are distinct machine states even when both produce the same output value. On the framework's reading,  $\perp$  is not a description waiting for someone

outside to press "run": if it runs at all, it runs itself (Path 1). That it does run is the occurrence commitment, which DA-1 uses and does not supply; given it, the machine has left  $c_0$  for  $c_1$ , by the definition of the first running configuration (D7). ZP-K carries  $c_0$ 's computational face as a KleeneStructure requirement (botCode\_is\_quine); reading  $c_0$  as the Kleene quine is the KleeneStructure commitment, not a theorem. What T-COMP proves is narrower: three of the four descriptions of  $\perp$  are equivalent to one another. It says nothing about external agents, and nothing about execution.

### Real-world analogy — The light that sees itself

Consider a camera that, instead of photographing external scenes, photographs only its own sensor. The image it produces is the state of the sensor; the sensor's state is the image. There is no external scene being captured — the camera IS the scene. On the framework's reading,  $\perp$  as a Kleene fixed point has exactly this structure: the program that runs is the program that describes what runs, so if it runs, nothing outside it is needed to run it. The analogy stops there: the same camera, wired the same way, can also sit switched off. That  $\perp$  does run is the framework's occurrence commitment, not something the self-reference supplies.

## A Note on Proof Purity

The computability machinery in ZP-K (Kleene's theorem, Rogers' fixed-point theorem) requires classical logic, inherited from Mathlib rather than a novel Zero Paradox commitment. ZP-K Section IV tabulates the measured axiom footprints. What is free of it is the choice-free CORE, and that is narrower than any layer name: ZeroParadox/AxiomProfile.lean §I is where the core is listed and each footprint measured — T-SNAP, the lattice, the Quine atom. Results elsewhere in the same layers do carry the dependency: decoration\_unique, the AFA decoration-uniqueness theorem in ZeroParadox/Settheory/APG.lean, is a ZP-J result and reports the full triple. Check the artifact for a given result rather than reading a layer name as a footprint.

This is analogous to a proof that invokes the intermediate value theorem. Reaching for a result from the standard, classically built libraries does not make your proof "non-constructive" in any meaningful sense — it means you are working in the standard mathematical setting. ZP-K's classical footprint is of the same character.

Remember: in ZP-K, DA-1 Path 1's Lean counterpart is da1\_closed\_concrete, which proves IsQuineAtom( $\perp$  : MachinePhase) and nothing computational; Path 3's Lean counterpart is the machinePhaseKleene instance's botCode\_is\_quine field, a KleeneStructure requirement that constant codes also meet, not a witness of execution and not a second independent proof. The two are carried together by da1\_paths\_unified as a conjunction; that they name one structural fact is the framework's reading. ZP-K does not claim to resolve the description-execution gap by philosophical argument.

## Self-Reference: Fixed Point vs. Oscillation

Not all self-reference is the same. The liar paradox — "this sentence is false" — produces a statement  $x$  with the property  $x = \text{NOT } x$ . In Boolean logic this has no fixed point: the sequence true, false, true, false, ... oscillates without resolving. This is the liar structure.

Gödel's incompleteness proof (1931) uses a different kind of self-reference. Using the diagonal lemma — a fixed-point theorem for formal languages — he constructed a sentence  $G$  satisfying  $G \leftrightarrow \neg \text{Prov}(G)$ : "this sentence is not provable in PA." The key move: he changed the predicate from "true" to "provable."

Provability is asymmetric in a way truth is not, so  $G$  does not oscillate. In a consistent system,  $G$  is true in the standard model of arithmetic but unprovable in PA — a fixed point, not an oscillation.

ZP-K's construction achieves the same structural form in the setting of  $\perp$ .  $\perp = \{\perp\}$  (the AFA Quine atom; `da1_closed_concrete` gives this for `MachinePhase`, where `selfMem` is defined as  $x = \perp$ , so the content is the uniqueness rather than the membership) is self-containing, not self-negating:  $x = f(x)$  where  $f$  is set-membership, not  $x = \text{NOT } x$ . The Kleene fixed point gives the same structure computationally: a code whose behavior is determined entirely by itself. Both witnesses resolve at a fixed point. Neither oscillates.

ZP-E's `t_snap_irreversible` goes further: it formally proves that the liar-type trajectory is structurally impossible in this framework, not merely absent. Once  $c_0$  transitions to  $c_1$ , the semilattice axioms guarantee no path returns. The sequence  $c_0 \rightarrow c_1 \rightarrow c_0 \rightarrow \dots$  cannot occur — the algebra forbids it.

This is a structural observation, not a claim that ZP proves Gödel's theorems. Gödel applied the diagonal lemma in formal arithmetic; ZP-K's construction achieves the same structural form —  $x = f(x)$  rather than  $x = \neg x$  — for the bottom element  $\perp$ . Both constructions use  $x = f(x)$  rather than  $x = \neg x$ , and both yield a stable fixed point rather than an oscillating sequence.

#### Key Result — ZP-K

T-COMP:  $\text{IsQuineAtom}(q) \leftrightarrow q = \perp \wedge (\forall x, q \vee x = x)$ . THREE characterisations are proved equivalent - self-containment (AFA), order, and algebra. The computational face is NOT a fourth clause of this theorem: it enters as an assumption, the KleeneStructure field `botCode_is_quine`. `da1_closed_concrete : IsQuineAtom( $\perp$  : MachinePhase)` proves the structural half only - it mentions no code and no execution. DA-1 Path 1 (structural): its Lean counterpart is proved, and Path 1 rules out an external executor, not an inert  $\perp$ . Path 3 (computational) shows that executing is not derivable from incompressibility; its Lean counterpart is a class field, assumed rather than derived.